Final Project Proposal: Audio Feedback Avoidance Agent

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1. Goal

Our goal is to create an audio feedback-avoidance agent for live music performance. This agent's algorithm will receive the individual raw audio frequencies and EQ parameters (volume, compression, etc.) from the various inputs (microphones, instruments, etc.) that make up a live performance over a short window of time. Using each input's audio frequencies, the algorithm will predict the likelihood of a potential feedback loop forming. It will make quick decisions based on its limited knowledge of audio frequencies on how to adjust the EQ parameters, especially seeking to avoid "feedback," which is the undesired, loud consequence of amplifier-microphone interactions.

Research exists in the field of feedback detection, but we haven't found any research that models the problem of feedback management and volume mixing as a Markov Decision Process (Mounir et al., 2025). In general, one-shot solutions seem to be more prevalent in the automated mixing and mastering space, which achieve great results but are less applicable to live music than recorded music; we hope that using a POMDP will allow us to create a more autonomous, live system.

2. Decision Making

Typically, live band music is played into a number of inputs (to simplify, imagine each instrument playing into its own microphone), and these inputs are routed to a mixing board, which gives a sound engineer the ability to control the volume and equalization levels of the different inputs in order to create the best sound for the audience. The engineer must make sequential decisions, adjusting the controls in response to both intended and unintended changes in the volumes of and interactions between inputs. They can also attempt to predict whether, given the structure of the song, a chorus or a breakdown is coming, where they might need to adjust the volume levels given a predicted change in the inputs. Specifically for live music, prevention of audio feedback requires a sound engineer to quickly know which of the various audio sources is involved in the feedback loop (which could be singular or multiple).

Despite these complexities, being a sound engineer is as much an art as it is a science. It's important for there to be clear sound at all times, but distorting the sound in order to achieve this clarity is not always the answer. A sound engineer must have the correct taste for what range of sound effects is acceptable for certain musicians in order to not disrupt the intended sound. Factoring these variables into the decision making process of a sound engineer illustrates how live mixing is quite analogous to other decision making problems we've explored in this class.

3. Sources of Uncertainty

Many sources of uncertainty exist in this problem. One is that feedback is unexpected and has various causes. An engineer wouldn't know when a feedback loop might kick in, and it's imperative to figure out where it's coming from and react quickly. For example, a singular feedback loop could have audio circle between a single microphone and its speaker output whereas a more complicated loop could involve multiple microphones and their individual speaker outputs. Ideally, in order to prevent these loops before they are audible to an audience, our algorithm must sense the loop beginning simply based on each audio input's relationship with its output along with the output of all of the other audio outputs. Whether or not the loop will grow to become a disturbance to the overall mix is uncertain, so it is up to the algorithm to make that decision based on its limited observations.

Beyond that, everything live musicians do is somewhat unexpected from the perspective of a sound engineer. For example, some musicians may start playing too loudly or too quietly for the ideal volume balance of the entire band, or a singer may step back from the microphone. One method to account for this source of uncertainty is for a sound engineer to understand the structural aspect of the music. The job of the mixing engineer is to account for these unpredictable inconsistencies.

4. Sketch of Solution

We plan to build our system in phases so that each phase produces a complete and defendable result on its own. By using milestones, the project does not depend on the full multi-input system working. We have not fully committed to this framework, and would greatly appreciate feedback. First, we would represent each audio source as an input channel with static metadata set at soundcheck (type, routing to speakers, and a boolean variable loopPotential) and observations computed for every step from its raw input audio signal. The loopPotential variable simply determines whether an input could possibly be involved in a feedback loop.

Phase 1: A singular loop decision process, one vocal mic and one speaker, modeled as a POMDP. The hidden state is the loop's stability margin (safe, near limit, ringing or howling). At each step, the algorithm outputs a belief, a probability distribution over these four states, determining its next actions. The observations are howling (feedback noise) features computed from the mic's spectrogram (Mounir et al., 2025). The actions are to raise or lower the channel's fader (volume) to the speaker, or hold. The reward favors staying near the target level, charges a small cost for every notch, and greatly penalizes howling. The agent tracks a belief over the hidden state and plans over it using QMDP to prioritize speed. We will simulate the loop with room impulse responses, building on Mounir et al.'s public code.

Phase 2: Multiple channel interaction. With several mics and speakers, feedback paths add together, so the algorithm will need to infer which loop is unstable before acting. To do this, it also observes the combined audio output and uses each channel's signal to determine the most likely source(s) of a potential feedback loop.

AI Use Disclosure

For this proposal, we used AI to help us plan and scope our approach. All research was done manually and writing in our own words, except for certain formatting in Latex.

References

Mina Mounir, Giuliano Bernardi, and Toon van Waterschoot. Robust and early howling detection based on a sparsity measure. *EURASIP Journal on Audio, Speech, and Music Processing*, 2025(14), 2025. doi: 10.1186/s13636-025-00399-1.